

tf.compat.v1.nn.dilation2d Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/dilation2d

Computes the grayscale dilation of 4-D input and 3-D filter tensors.

Function Signatures

```
tf.compat.v1.nn.dilation2d( input, filter=None, strides=None,
rates=None, padding=None, name=None, filters=None,
dilations=None )
```

Code Examples

```
tf.compat.v1.nn.dilation2d(
input,
filter=None,
strides=None,
rates=None,
padding=None,
name=None,
filters=None,
dilations=None
)
```

```
tf.compat.v1.nn.dilation2d(  
input,  
filter=None,  
strides=None,  
rates=None,  
padding=None,  
name=None,  
filters=None,  
dilations=None  
)
```

```
output[b, y, x, c] =  
max_{dy, dx} input[b,  
strides[1] * y + rates[1] * dy,  
strides[2] * x + rates[2] * dx,  
c] +  
filter[dy, dx, c]
```

```
output[b, y, x, c] =  
max_{dy, dx} input[b,  
strides[1] * y + rates[1] * dy,  
strides[2] * x + rates[2] * dx,  
c] +  
filter[dy, dx, c]
```

Documentation

Computes the grayscale dilation of 4-D input and 3-D filter tensors.

The input tensor has shape `[batch, in_height, in_width, depth]` and the filter tensor has shape `[filter_height, filter_width, depth]`, i.e., each input channel is processed independently of the others with its own structuring function. The output tensor has shape `[batch, out_height, out_width, depth]`. The spatial dimensions of the output tensor depend on the padding algorithm. We currently only support the default "NHWC" data_format.

In detail, the grayscale morphological 2-D dilation is the max-sum correlation (for consistency with `conv2d`, we use unmirrored filters):

Max-pooling is a special case when the filter has size equal to the pooling kernel size and contains all zeros.

Note on duality: The dilation of input by the filter is equal to the negation of the erosion of -input by the reflected filter.

Returns